



RAMCO INSTITUTE OF TECHNOLOGY

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Department of Computer Science and Business System

Academic Year: 2024- 2025(Odd/Even Semester)

Degree, Semester & Branch: B.Tech., V & CSBS

Course Code & Title: CW3551 & Data and Information Security

Name of the Faculty member (s): Ms.M. Preethi Ram, AP/CSBS

Innovative Practice Description

- **Unit / Topic:** Unit 1 / The SDLC
- **Course Outcome:** CO1
- **Topic Learning Outcome:** TLO 1,2,3
- **Activity Chosen:** Flipped classroom

Justification: Flipping the classroom is an approach that reverses the conventional teaching method. In this method, the traditional in-person teaching is restructured to cater to the unique learning requirements of individual students. Students take charge of their learning by independently studying course materials before attending class, such as reading materials and pre-recorded video lectures. This approach allows educators to reimagine in-class activities, which may include interactive problem-solving sessions, aimed at keeping students actively involved in the subject matter.

Time Allotted for the Activity: 30 Minutes

- **Details of the Implementation:**

- The learning materials were posted to the students in canvas two days before the topic discussion.
- The students are instructed to study the material and take notes on the topic based on their understanding.
- On the day of the activity the students were grouped into a team of 2 members and they are asked to discuss about the topic they learnt in their home through the study materials for the duration of 10 minutes.
- Based on the study material they have gone through, the SDLC models were given for each group separately and by discussing with their team members they have clear idea of it.
- After the discussion one member from each team had a chance to present and share the way of methods to the entire class as shown in the Fig.1 and each student created a flowchart based on the real time example as shown in Fig.2 & 3.

- **CO – PO / PSO mapping:**

CO	PO1	PO2	PO3	PO4	PO6	PO7	PO9	PSO1
CO1	3	3	2	2	2	1	1	3

PO mapped:

Innovative practice	PO1 3	PO2 3	PO3 2	PO4 2	PO6 2	PO7 1	PO9 1
Justification for correlation	Students will acquire in-depth knowledge about the models of SDLC	Students will analyze given use cases and scenarios using engineering sciences.	Students will design and develop SDLC model-based solutions.	Students will investigate complex problems related SDLC models, exploring potential improvements and innovative approaches.	Students will become proficient in using tools for software development, such as version control systems, integrated development environments (IDEs), and testing frameworks.	Students will learn to design software solutions that are efficient and sustainable, considering factors like energy consumption and the lifecycle impact of the software.	Students will develop skills to function effectively as part of a team, understanding roles, responsibilities, and the importance of teamwork in successful software projects.

PSO Mapped:

Innovative practice	PSO1 3
Justification for correlation	Students will learn how to incorporate advanced technologies into each phase of the SDLC. For instance, they might use AI/ML in the design and development phases or leverage Cloud Computing for deployment and maintenance.

Images / Screenshot of the practice:

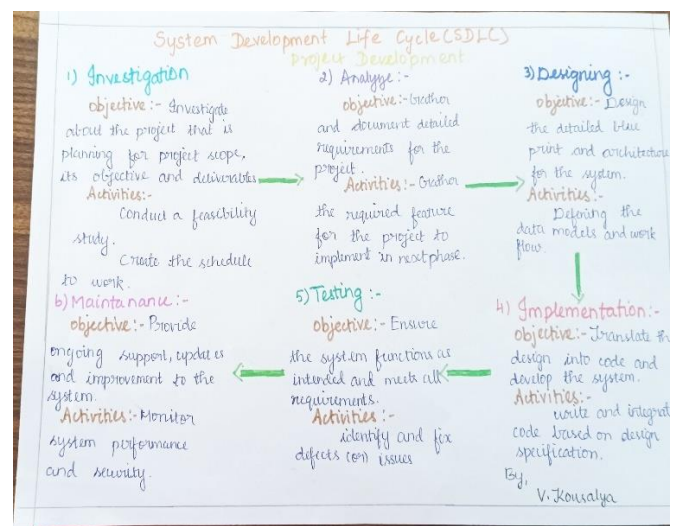
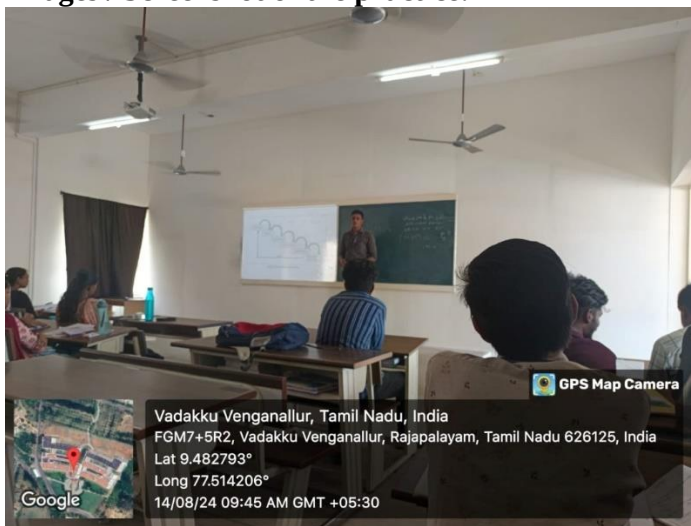


Fig 1: Presentation done by Hari Krishna R

Fig 2: Flowchart done by V.Kousalya with her own example. (Ex: For Project development in company)

Reflective Critique:

- **Feedback of practice from students and other stakeholders:**
 - ✓ Student felt good, since they can study at their own pace/time.
 - ✓ They felt that through such learning, they can explore more.
- **Benefit of the practice:**
 - ✓ More one-to-one time between teacher and student.
 - ✓ More collaboration time for students.
 - ✓ Students learn at their own pace.
 - ✓ Practical things – like missing class due to illness – become less problematic.
 - ✓ It encourages students to come to class prepared.
- **Challenges faced in implementation:**
 - ✓ Relies on student preparation – few students did not come prepared.
 - ✓ The depth of the subject can be dictated by the student themselves or the group the student is working with.
 - ✓ The time and effort required from a teacher's perspective initially when creating the flipped class material is higher than for a traditional class.

References:

- [https://www.teachthought.com/learning/the-definition-of-theflippedclassroom/#:~:text=A%0flipped%20classroom%20is%20a,the%20students%20independently%20at%20home.](https://www.teachthought.com/learning/the-definition-of-theflippedclassroom/#:~:text=A%0flipped%20classroom%20is%20a,the%20students%20in%20independently%20at%20home.)
- https://en.wikipedia.org/wiki/Flipped_classroom
- [youtube.com/watch?v=BCIxikOq73Q](https://www.youtube.com/watch?v=BCIxikOq73Q)
- <https://omerad.msu.edu/teaching/teaching-skills-strategies/27-teaching/162-whatwhy-and-how-to-implement-a-flipped-classroom-model>

Signature of Faculty Member

HOD